

Rajender Karra

AI-Augmented Software Engineer

18 Years | Spring Boot | Microservices & GCP | Building Scalable Cloud-Native Enterprise Applications | Solution Architecture | Hyderabad, India | +91-8179020343 | karra.rajender@gmail.com | [LinkedIn](#)

Executive Profile

AI-Augmented Software Engineer with 18 years of enterprise product development experience across Banking and Insurance domains. Expertise in Java 21, Spring Boot 3.x, Microservices, PostgreSQL, Redis, Kubernetes and Google Cloud Platform (GCP). Experienced in designing APIs, database schemas and microservice contracts, leading engineering teams, optimizing performance and leveraging GitHub Copilot for AI-assisted software development. Passionate about Solution Architecture, Distributed Systems and modern AI engineering.

Engineering Achievements

- 18 years of enterprise product engineering experience across Banking and Insurance domains.
- Contributed to the greenfield development of HSBC's next-generation cloud-native collections platform, modernizing legacy collections capabilities through scalable microservices architecture.
- Owned the end-to-end design, development, and deployment of 20+ Spring Boot 3.x cloud-native microservices, delivering scalable and production-ready enterprise services.
- Designed enterprise REST APIs, database schemas, and microservice contracts by translating business requirements from Jira user stories and Figma designs into scalable technical solutions.
- Provided technical leadership to a team of 10 software engineers, driving solution design, code quality, and successful production deliveries.
- Improved Account Aggregate API response time by 80% (10s → 2s) through parallel service execution.
- Improved API implementation productivity by approximately 30% through AI-assisted development using GitHub Copilot.
- Implemented Redis-based in-memory caching for configuration data to improve application performance and reduce database access.
- Achieved 95%+ JUnit test coverage across developed services, improving code quality and release confidence.
- Delivered multiple production releases aligned with enterprise banking quality, security, and compliance standards.

Core Competencies

Enterprise Product Development • Solution Design & Architecture • Cloud-Native Application Development • Enterprise API Design & Development • Microservices Architecture • Distributed Systems • Database Design & Data Modeling • Performance Engineering • Secure API Development • Technical Leadership • AI-Augmented Software Engineering • High-Level Design • Low-Level Design • Agile Software Delivery

Technology Stack

Category	Skills
Languages	Java 21, SQL, Python
Frameworks	Spring Boot 3.x, Spring, Hibernate, LangChain, LangGraph
Architecture	Microservices, REST APIs, OpenAPI, Distributed Systems, RAG
Security	Spring Security, OAuth2, JWT
Cloud	GCP, AWS
Containers	Kubernetes, Docker
Data	PostgreSQL, Oracle, SQL Server, Redis, Liquibase
Testing	JUnit5, Mockito
Observability	Open Telemetry, Prometheus, Micrometer
Code Quality	SonarQube, Sonatype, Checkmarx, Cyberflow
AI	GitHub Copilot, AI Agents, MCP (Learning), Prompt Engineering

Professional Experience

LTIMindtree | Associate Principal Software Engineer | Jun 2021 - Present

Project: Next Generation Collections Platform (C48)

Client: HSBC Software Development (India) Pvt. Ltd.

Domain: Banking – Collections & Recovery

Project Overview: Contributed to HSBC's strategic greenfield initiative to modernize the global collections platform by replacing legacy systems with a scalable, cloud-native microservices architecture on Google Cloud Platform (GCP).

Technology Stack: Java 21, Spring Boot 3.x, Spring Data JPA, PostgreSQL, Redis, GCP, Pub/Sub, Kubernetes, Docker, OpenAPI, JWT, Liquibase, JUnit5, Mockito, Open Telemetry, Micrometer, Prometheus, GitHub, Jenkins, Maven, SonarQube, Sonatype, Checkmarx, Cyberflow, Jira, Confluence, GitHub Copilot.

Key Contributions

- Led the end-to-end delivery of enterprise REST APIs, from solution design and technical estimation through implementation, testing, production deployment, and post-production support.
- Designed enterprise REST APIs, PostgreSQL database schemas, and microservice contracts by translating business requirements from Jira user stories and Figma UI/UX designs into scalable technical solutions.
- Designed, developed, and deployed 20+ Spring Boot 3.x cloud-native microservices supporting HSBC's next-generation collections platform.
- Contributed to the greenfield development of HSBC's cloud-native collections platform, helping modernize legacy collections capabilities through scalable microservices architecture.
- Improved the Account Aggregate API response time by 80% (approximately 10 seconds to 2 seconds) by redesigning service orchestration using parallel processing.
- Increased API implementation productivity by approximately 30% through effective adoption of GitHub Copilot (Agent Mode) in the software development lifecycle.

- Implemented Redis-based in-memory caching for administrative configuration data, improving application performance and reducing database dependency.
- Provided technical leadership to a team of 10 software engineers, driving solution design, code reviews, mentoring, sprint execution, and successful production deliveries.
- Participated in High-Level Design (HLD), Low-Level Design (LLD), backlog refinement, technical estimations, and architecture discussions with cross-functional stakeholders.
- Achieved 95%+ JUnit test coverage using JUnit 5 and Mockito, improving code quality, release confidence, and maintainability.
- Integrated quality and security gates using SonarQube, Sonatype Nexus IQ, Checkmarx, and Cyberflow (SAST, Container Security, and DAST) to ensure compliance with enterprise banking standards.
- Delivered multiple production releases while adhering to HSBC's banking quality, security, and compliance standards.

Project: Mortgage Process Optimization (MPO)

Domain: Banking – Mortgage Processing

Client: HSBC Software Development (India) Pvt. Ltd.

Project Overview: Contributed to the modernization of HSBC's mortgage processing platform by designing and implementing cloud-native microservices to streamline mortgage workflows, improve scalability, and support enterprise banking operations.

Technology Stack: Java 21, Spring Boot 3.x, PostgreSQL, Kubernetes, Docker, AWS cloud, PostgreSQL, JWT, JUnit5, Mockito, GitHub, Jenkins, Maven, SonarQube, Sonatype, Checkmarx, Cyberflow, Jira, Confluence.

Key Contributions

- Designed enterprise REST APIs, PostgreSQL database schemas, and microservice contracts based on business requirements and functional specifications.
- Developed cloud-native Spring Boot microservices supporting mortgage processing workflows using Java 21 and PostgreSQL.
- Participated in High-Level Design (HLD), Low-Level Design (LLD), backlog refinement, technical estimation, and sprint planning activities.
- Collaborated closely with business stakeholders, product owners, QA, and DevOps teams to deliver scalable, production-ready banking solutions.
- Followed enterprise engineering standards by implementing comprehensive unit testing using JUnit 5 and Mockito while maintaining code quality through SonarQube and security validation using Checkmarx and Cyberflow.
- Supported multiple production releases, ensuring successful deployment, post-production stabilization, and adherence to HSBC's banking quality and security standards.

EbixCash | Technical Lead | Jan 2013 - Jun 2021

Product: The Policy Processor

Domain: Insurance

Key Customers: General Reinsurance Corporation, The Allstate Corporation, Nationwide Mutual Insurance

Project Overview: The Policy Processor is an enterprise insurance platform used by global insurance organizations to automate policy administration, premium processing, business rule execution, and integration with downstream insurance systems.

Technology Stack: Java, Spring, Spring Boot, Microservices, JMS, SQL Server, Git, Jenkins, Maven, WildFly.

Key Contributions

- Led a cross-functional engineering team responsible for designing, developing, and enhancing enterprise insurance product capabilities for global customers.
- Contributed to solution design, application modernization, and migration of legacy components to Spring Boot and Microservices-based architecture.
- Designed and implemented business-critical features by collaborating with product managers, business analysts, and QA teams to translate functional requirements into scalable technical solutions.
- Conducted technical design reviews, code reviews, and mentoring sessions to improve code quality, engineering standards, and team productivity.
- Drove end-to-end feature delivery, including requirements analysis, technical design, development, testing, release management, and production support.
- Collaborated with cross-functional teams to deliver multiple product releases while ensuring application stability, maintainability, and performance.

Oracle Financial Services | Principal Consultant | Jul 2008 - Jan 2013

Product: Oracle FLEXCUBE Direct Banking

Domain: Internet Banking

Key Customers: International Bank of Qatar (IBQ), Halifax Bank of Scotland (HBoS), Federal Bank of Middle East

Project Overview: Oracle FLEXCUBE Direct Banking is a globally recognized enterprise digital banking platform that enables financial institutions to deliver secure retail and corporate internet banking services. Contributed to the implementation and customization of banking solutions for international banking clients.

Technology Stack: J2SE 5.0, EJB, XSL, XML, Web Services, SOA, Oracle FLEXCUBE Direct Banking (FCDB) Framework, IBM MQ, IBM WebSphere Application Server 7.1, Oracle 11g, Eclipse, HTML and Java script.

Key Contributions

- Designed and developed enterprise banking functionalities for Oracle FLEXCUBE Direct Banking, supporting retail and corporate internet banking solutions.
- Collaborated with solution architects to implement new business capabilities and enhance existing banking modules while adhering to enterprise architecture standards.
- Customized and integrated banking applications with third-party systems using Web Services, IBM MQ, and SOA-based integration patterns.
- Collaborated with solution architects, international banking clients, and cross-functional teams to translate business requirements into customized enterprise banking solutions and successfully deliver customer implementations.
- Participated in the complete software development lifecycle, including requirements analysis, solution design, development, testing, implementation, production support, and customer deployments.

Education

B.Tech, Kakatiya Institute of Technology and Science (KITS), Kakatiya University